OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 22 OF 55

AI-Based Customer Complaint Classification and Priority Prediction Using NLP

Domain: NLP & Customer Experience Automation | **Module:** project_22_complaint_classification

OFFICIAL ABSTRACT

Customer complaints provide valuable information about problems faced by users, but handling a large number of complaints manually can be time-consuming and inefficient. Complaints received through emails, online forms, chat platforms, and customer-support systems may contain different types of issues, making it difficult for support teams to quickly identify which complaints require immediate attention. The proposed AI-Based Customer Complaint Classification and Priority Prediction System Using Natural Language Processing (NLP) is designed to automatically understand, categorize, and prioritize customer complaints. The proposed system is developed using Python, NLP techniques, and machine learning algorithms to analyze the textual content of complaints. The system processes the complaint, identifies important textual patterns, and classifies it into relevant categories such as payment issues, product complaints, delivery problems, technical issues, account-related problems, or service dissatisfaction. This reduces the need for support staff to manually read and categorize every complaint. A major uniqueness of the proposed system is the combination of complaint classification and priority prediction in a single intelligent workflow. After identifying the complaint category, the system analyzes the seriousness and potential impact of the issue to predict an appropriate priority level such as low, medium, high, or critical. For example, a general product inquiry may receive a lower priority, while a complaint involving failed payment, account access, or a service interruption can be identified as requiring faster attention. The system can also identify important words and contextual patterns within complaints instead of depending only on predefined keywords. This allows similar complaints to be classified even when customers describe the same problem using different words or sentences. The model can be trained using historical complaint data so that its classification and priority prediction capabilities can improve based on previously handled cases. Another useful feature is the generation of a structured complaint summary, which can help support teams quickly understand the issue without reading

the complete customer message. The system can display the complaint category, predicted priority, and relevant information in an organized format. With further development, it can be integrated with customer-support platforms to analyze incoming complaints automatically and support real-time complaint routing. The proposed system can be applied in e-commerce, banking, telecommunications, education, healthcare services, and other organizations that manage large volumes of customer feedback. By automatically identifying both what the complaint is about and how urgently it should be handled, the system can help organizations reduce response time and improve customer-service efficiency. Overall, this project demonstrates how NLP and machine learning can convert unstructured customer complaints into actionable support information. The combination of automated classification, priority prediction, contextual analysis, and structured complaint handling provides a practical AI-based approach for building faster, smarter, and more efficient customer-support systems.

📌 Project Summary & System Specifications

Project Title	AI-Based Customer Complaint Classification and Priority Prediction Using NLP
Specialization / Category	NLP & Customer Experience Automation
Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5022`
Local Implementation	`project_22_complaint_classification`